

# ANANTH JONNAVITTULA

San Francisco, CA

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[ananth.fyi](https://www.ananth.fyi) [Google Scholar](https://scholar.google.com/citations?user=...)

## Summary

Robotics researcher with 8+ years of experience across academic and industrial domains, specializing in perception-driven manipulation, visual imitation learning, and generative AI for robotics. Proven track record deploying production-grade vision systems and publishing impactful research in reinforcement and imitation learning. Seeking to develop on-foundational AI/ML technologies.

## Technical Skills

**Languages:** Python, URScript, KUKA KRL

**Frameworks/Tools:** PyTorch, TensorFlow, OpenCV, ROS, Isaac Sim, Blender, TorchScript, GitHub

**Robotics Platforms:** Universal Robots, ABB, FANUC, KUKA, FrankaEmika, Fetch

**Research Interests:** Visual Imitation Learning, Robot Perception, Self-Supervised Learning, Human-Robot Interaction, Foundation Models

## Education

### Virginia Tech

*PhD in Mechanical Engineering*

**Aug 2020 – Aug 2024**

*Blacksburg, VA*

### Worcester Polytechnic Institute

*MS in Robotics Engineering*

**Aug 2015 – May 2017**

*Worcester, MA*

### SASTRA University

*B.Tech in Electronics and Instrumentation*

**Jun 2011 – May 2015**

*Tanjore, India*

## Work Experience

### Senior Research Engineer

*Samsung Research America*

**Sep 2024 – Present**

*Mountain View, CA*

- Developed VR based teleoperation systems for humanoid robots
- Designed universal teleoperation system stack for AI data collection
- Developed Nav2 based navigation for humanoid robots

### AI Researcher

*Foundation Future Industries Inc*

**Dec 2024 – Sep 2025**

*San Francisco, CA*

- Developed Diffusion Policy based Imitation Learning policies for dexterous manipulation
- Created segmentation and pose estimation pipelines using DINOv2 and SAM2 for uncommon automotive parts
- Integrated NeRF-based SLAM pipelines for scene understanding in real world factories
- Engineered Nvidia Isaac Sim based simulation environment for custom humanoid robot

### Graduate Research Assistant

*Virginia Tech*

**Aug 2020 – Aug 2024**

*Blacksburg, VA*

- Developed an imitation learning method that can learn long horizon tasks from a single video demonstration
- Implemented Soft Actor-Critic based Reinforcement Learning approaches for manipulation of YCB objects
- Engineered an inverse reinforcement learning based algorithm to learn from imperfect user demonstrations
- Spearheaded the development of a variational autoencoder based algorithm for imitation learning
- Conducted stability analysis for imitation learning methods in the context of shared autonomy

## Research and Development Intern

Jun 2022 – Aug 2022

ABB Inc

San Jose, CA

- Implemented and analyzed methods that use stereo vision or RGBD images for instance segmentation
- Architected an instance segmentation algorithm for small parcels singulation using detectron2
- Devised a BlenderProc based pipeline to generate synthetic datasets for training models
- Engineered a pipeline to export trained models from Python to C++ using Torchscript

## Robotics/Vision Engineer

Jun 2017 – May 2020

Parker Hannifin Corp.

Haverhill, MA

- Engineered an automated cell for palletizing over 100 different SKUs to reduce labor costs
- Designed and implemented an automated laser marker that doubled throughput in multiple manufacturing cells
- Developed a robotic cell using UR5 robot and cameras for part recognition and end capped filter
- Engineered and deployed a urethane end capping cell using FANUC robots

## Engineering Intern

Jan 2017 – Apr 2017

Parker Hannifin Corp.

Haverhill, MA

- Conducted feasibility analysis on automated end capping using collaborative robots
- Programmed controllers for automated part feeding using vibratory feeders
- Implemented image recognition using Keyence CV-X series to detect orientation of parts

## Graduate Research Assistant

Jan 2016 – May 2016

Worcester Polytechnic Institute

Worcester, MA

- Conducted analysis for range of motion and kinematic requirements for a 2 DOF hydro-muscle actuated leg
- Designed a coupling mechanism that locks the leg while maintaining pose in case of serious failure
- Orchestrated the development closed-loop control system for leg actuation using four pairs of hydro-muscles
- Developed controllers for hydraulics and coupling mechanism using Arduino microcontrollers

## Publications

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### VIEW: Visual Imitation Learning with Waypoints

2025

*Springer Autonomous Robots*

Ananth Jonnavittula, Sagar Parekh, and Dylan P. Losey

### SARI: Shared Autonomy across Repeated Interaction

2024

*ACM Transactions on Human-Robot Interaction*

Ananth Jonnavittula, Shaunak A. Mehta, and Dylan P. Losey

### Communicating Robot Conventions through Shared Autonomy

2022

*International Conference on Robotics and Automation (ICRA)*

Ananth Jonnavittula and Dylan P. Losey

### Here's What I've Learned: Asking Questions that Reveal Reward Learning

2022

*ACM Transactions on Human-Robot Interaction*

Soheil Habibian, Ananth Jonnavittula, and Dylan P. Losey

### Learning to Share Autonomy Across Repeated Interaction

2021

*IEEE International Conference on Intelligent Robots and Systems (IROS)*

Best Student Paper Finalist

Ananth Jonnavittula and Dylan P. Losey

### I know what you meant: Learning human objectives by (under)estimating their choice set

2021

*IEEE International Conference on Robotics and Automation (ICRA)*

Ananth Jonnavittula and Dylan P. Losey

## Patents

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### Biologically inspired joints and systems and methods of use thereof

2020

US010632626B2